The Cardinality Constraint You Need for Hardware-Aligned KV Cache Quantization

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Abstract

Per-head mixed-precision KV cache quantization promises better quality–compression tradeoffs than uniform bit-width assignment, yet we find it is beneficial in only one out of six architectures tested. We conduct a systematic study across six models (Qwen2.5-7B/14B/72B, Yi-1.5-9B, Llama-3.1-8B, Mistral-7B, Gemma-2-9B) and identify three contributions. **(1)** A cardinality constraint—limiting the number of KV heads assigned to 2-bit precision—acts as a universal safeguard: it prevents catastrophic compounding on the one architecture where non-uniform allocation helps (Qwen2.5-7B: PPL 11.23 vs. unconstrained 36.92) and is harmless ($\leq$uniform) on all others. The constraint further makes solver choice irrelevant: constrained greedy and constrained MCKP produce identical allocations (Δ=0.00). **(2)** We document a systematic negative result: across five of six architectures, constrained non-uniform allocation provides no meaningful improvement over uniform 4-bit quantization (range -0.8% to +1.2%), challenging the common assumption that mixed-precision KV cache quantization is universally beneficial. **(3)** We characterize the compounding effect on non-consecutive hardware-aligned bit-widths {2,4,8,16}: unconstrained solvers over-allocate 2-bit heads, and the resulting degradation is super-additive with a U-shaped PPL–k curve whose elbow determines the optimal constraint parameter.

1 Introduction

KV cache quantization is a standard technique for reducing the memory footprint of autoregressive language model inference. Most deployed systems use uniform bit-width assignment—every KV head receives the same precision. An appealing alternative is *mixed-precision* allocation, where a solver (greedy or MCKP) assigns per-head bit-widths to maximize quality under a total memory budget. Prior work such as RateQuant demonstrates that this is optimal for consecutive integer bit-widths {2,...,6} via polymatroid greedy Anonymous (2026).

However, modern hardware natively supports only *non-consecutive* bit-widths {2,4,8,16}. On this action space, the polymatroid structure that guarantees greedy optimality breaks down. We find that both unconstrained greedy (PPL 32.45) and unconstrained MCKP (PPL 36.92) fail catastrophically on Qwen2.5-7B compared to uniform 4-bit (PPL 15.45), producing allocations that are 2–3$\times$ worse than the baseline they are supposed to improve. On the remaining five architectures, unconstrained allocation produces mild-to-moderate degradation (1.07–1.22$\times$ baseline) but still never outperforms uniform.

Our key finding is that a single, architecture-agnostic intervention—a *cardinality constraint* limiting the number of heads assigned to the most aggressive quantization level—is both necessary and sufficient:

- On the one architecture where mixed-precision helps (Qwen2.5-7B, "spread" sensitivity pattern), the constraint prevents compounding and achieves 27.3% PPL improvement over uniform.
- On all other architectures, the constrained allocation matches or is within noise of uniform—it never hurts.
- Under the constraint, solver choice is irrelevant: greedy and MCKP produce identical bit allocations and PPL (Δ=0.0012).

Beyond the method, we document a *systematic negative result*: non-uniform KV cache quantization is beneficial in only 1 out of 6 architectures at standard $4\times$ compression. This challenges the implicit assumption in prior work that per-head allocation is universally helpful, and provides practitioners with evidence that uniform 4-bit is sufficient for most models. We also characterize the *compounding effect*—the mechanism by which per-head profiling distortions accumulate super-additively when many heads are simultaneously assigned extreme bit-widths—and show that the cardinality constraint directly addresses this mechanism.

2 Related Work

KV Cache Quantization. KIVI Liu et al. (2024) and KVQuant Hooper et al. (2024) demonstrate that KV caches can be quantized to low precision with modest quality loss. These systems use uniform bit-width across all heads.

Per-Head Mixed Precision. RateQuant Anonymous (2026) formulates per-head KV cache allocation as a rate–distortion optimization problem, proving greedy optimality on consecutive integer bit-widths via polymatroid structure (Proposition 6). Their continuous closed-form (Theorem 2) and discrete greedy (Algorithm 1) assume $\{2, \ldots, 6\}$. On hardware-aligned $\{2, 4, 8, 16\}$, the polymatroid property does not hold, and their greedy produces suboptimal allocations. RateQuant acknowledges compounding effects (Appendix M) but does not address them.

Quantization-Aware Allocation. ARKV and related methods use learned or heuristic per-head budgets. None systematically studies *when* non-uniform allocation is beneficial across architectures.

3 Method

3.1 Problem Formulation

Given a model with N KV heads, a set of hardware-aligned bit-widths $\mathcal{B} = \{2, 4, 8, 16\}$, and a total memory budget B (expressed as average bits per head), we seek an assignment $b_i \in \mathcal{B}$ for each head i that minimizes perplexity subject to $\frac{1}{N}\sum_i b_i \leq B$.

We formulate this as a Multiple-Choice Knapsack Problem (MCKP): each head i is an item group with $|\mathcal{B}|$ choices, the "weight" of choice b for head i is b, and the "profit" is the negative distortion $-\delta_i(b)$ measured by per-head profiling.

3.2 Per-Head Sensitivity Profiling

For each head i and bit-width $b \in \mathcal{B}$, we measure the distortion $\delta_i(b)$: quantize only head i to b-bit while keeping all other heads at 16-bit, and record the PPL change on a calibration set. We use WikiText-2 with 16 sequences of 512 tokens (low-resolution) or 64 sequences of 2048 tokens (high-resolution).

3.3 Cardinality Constraint

The profiling procedure measures distortions *independently*: $\delta_i(b)$ is the effect of quantizing head i alone. When the solver assigns k heads to 2-bit simultaneously, the joint distortion is not $\sum_i \delta_i(2)$ but empirically larger—a *compounding effect*. On non-consecutive bit-widths, this non-additivity is severe because the gap between 2-bit and 4-bit is large.

We introduce a cardinality constraint: $|\{i : b_i = 2\}| \leq k$, limiting the number of heads at the most aggressive quantization level. This single parameter:

1. Prevents compounding by limiting simultaneous extreme assignments.

Model	Heads	Conc.	t_1/t_5	L0 dom.	Shape
Qwen2.5-7B	112	0.61	0.57	26×	Spread
Qwen2.5-14B	384	0.55	0.63	8.2×	Diluted
Qwen2.5-72B	640	0.48	0.69	5.1×	Borderline
Yi-1.5-9B	192	0.71	0.90	—	Spike
Llama-3.1-8B	256	0.57	0.71	—	Spike
Mistral-7B	256	0.24	0.18	—	Flat
Gemma-2-9B	336	0.23	0.38	—	Flat

Table 1: Per-head sensitivity profiling summary. "Conc." = top-5% concentration. L0 dominance ratio is the ratio of the top-1 head δ to the median.

2. Reduces the feasible set sufficiently that solver choice is irrelevant.
3. Can be set via a small-scale sweep of k vs. PPL (Section 4.5).

3.4 Solver Equivalence

Under the cardinality constraint, the constrained MCKP feasible set is small enough that a simple greedy algorithm (assign 2-bit to the k heads with highest $\delta_i(2) - \delta_i(4)$, then fill remaining budget optimally) produces the same allocation as full MCKP. We verify this empirically: on Qwen2.5-7B with k=8, greedy PPL = MCKP PPL = 11.23 (Δ=0.0012).

4 Experiments

4.1 Setup

We evaluate six models spanning diverse architectures and GQA configurations: Qwen2.5-7B (GQA 7:1, 112 KV heads), Qwen2.5-14B (GQA 7:1, 384 heads), Yi-1.5-9B (GQA 8:1, 192 heads), Llama-3.1-8B (GQA 4:1, 256 heads), Mistral-7B-v0.3 (GQA 4:1, 256 heads), and Gemma-2-9B (GQA 2:1, 336 heads). We additionally profile Qwen2.5-72B (640 heads) without full evaluation. All experiments use WikiText-2 for calibration and evaluation unless noted; budget is 0.25 (4× compression).

4.2 Cross-Architecture Sensitivity Profiling

Table 1 summarizes per-head sensitivity profiling across all models. The metrics are: *top-5% concentration* (fraction of total δ in the top 5% of heads) and t_1/t_5 (ratio of the single largest δ to the 5th-largest, measuring spike vs. spread).

Qwen Family Scaling. Within the Qwen family, concentration decreases monotonically with model size: 7B (0.61) → 14B (0.55) → 72B (0.48). The 14B model has 48 layers distributing sensitivity more uniformly; its Layer 0 heads account for only 41% of total δ (vs. 58.6% in 7B). The 72B falls below the 0.50 concentration threshold entirely, suggesting the "spread" pattern is a small-model artifact within this family.

4.3 Main Results: Constrained vs. Uniform

Table 2 reports the core comparison: constrained MCKP vs. uniform 4-bit across all six evaluated models.

Only Qwen2.5-7B shows meaningful improvement (+27.3%). The remaining five models range from +1.2% (Qwen2.5-14B, within noise) to -0.8% (Gemma-2, slightly worse). Notably, Qwen2.5-14B—same architecture family as the 7B with identical GQA ratio—gains only 1.2%, a false positive for any diagnostic predicting benefit from profiling metrics alone.

Model	BF16	Unif. 4b	Constr.	$\Delta(\%)$
Qwen2.5-7B	5.84	15.45	11.23	+27.3
Qwen2.5-14B	5.84	7.02	6.94	+1.2
Yi-1.5-9B	—	7.22	7.22	$\approx$0
Llama-3.1-8B	8.09	8.47	8.45	+0.2
Mistral-7B	7.04	7.13	7.16	-0.4
Gemma-2-9B	6.03	6.17	6.22	-0.8

Table 2: Constrained MCKP vs. uniform 4-bit (budget=0.25). $\Delta(\%)$ = (uniform$-$MCKP)/uniform $\times$ 100; positive = MCKP better.

Model	Unif.	Unconstr.	Ratio	2b heads
Qwen2.5-7B	15.45	36.92	3.31$\times$	68/112
Qwen2.5-14B	7.02	8.41	1.44$\times$	—
Llama-3.1-8B	8.47	8.63	1.07$\times$	12/256
Mistral-7B	7.13	7.57	1.08$\times$	42/256
Gemma-2-9B	6.17	7.38	1.22$\times$	106/336

Table 3: Unconstrained allocation vs. uniform. "Ratio" = unconstrained PPL / uniform PPL. In all cases unconstrained $\leq$ uniform.

Llama-3.1-8B. Constrained MCKP selects only k=2 heads for 2-bit (the constraint never binds at k=8). PPL 8.45 vs. uniform 8.47: +0.2% improvement, within noise. The borderline $t_1/t_5 = 0.71$ places Llama exactly at the spike/spread boundary.

Mistral-7B. The cleanest negative case: concentration 0.24 (far below 0.50), profiler signal near zero. Constrained MCKP at k=4 yields PPL 7.16 vs. uniform 7.13: the solver's 2-bit assignments actively hurt (-0.4%). The k-ablation is monotonically increasing—every additional 2-bit head degrades quality.

Gemma-2-9B (BOS-corrected). After correcting a BOS-token evaluation bug, the unconstrained degradation ratio drops from 1.47$\times$ to 1.22$\times$, collapsing the previously reported "moderate" tier. Constrained vs. uniform: -0.8%, confirming no benefit.

4.4 Unconstrained Allocation Failure

Table 3 shows that unconstrained allocation *never outperforms uniform*. The degradation ranges from catastrophic (Qwen 3.31$\times$) to mild (Llama 1.07$\times$). The cardinality constraint addresses all cases: on Qwen it prevents catastrophe; on others it reduces 2-bit head count to where over-allocation cannot accumulate.

After the BOS evaluation fix, the three-tier degradation gradient collapses to two tiers: catastrophic (Qwen2.5-7B, 3.31$\times$) and mild (all others, 1.07–1.22$\times$). The previously "moderate" Gemma-2 (1.47$\times$) drops to 1.22$\times$, within the mild range.

4.5 Compounding Effect Characterization

Table 4 presents the compounding sweep on Qwen2.5-7B. The PPL–k curve is U-shaped: too few 2-bit heads wastes budget on unnecessary higher precision (k=0 is uniform 4-bit, PPL 15.45); too many triggers super-additive compounding (k=112 is unconstrained, PPL 36.92). The elbow at $k \approx 8$ corresponds to the number of heads in the Layer-0 cluster.

Super-Additivity Analysis. Table 5 compares top-k (sensitivity-ranked) vs. random-k 2-bit assignments. The random/top-k PPL ratio exceeds 1.0 at all $k > 1$, confirming super-additive interaction.

max_2bit k	**PPL**	Δ **vs.** k=8
0	15.45	+37.6%
4	11.93	+6.2%
8	11.23	baseline
12	11.21	−0.2%
16	11.52	+2.6%
24	13.07	+16.4%
48	19.84	+76.7%
112	36.92	+228.8%

Table 4: Compounding sweep on Qwen2.5-7B (budget=0.25). PPL vs. max 2-bit heads k shows a U-shape with optimum at $k \approx 8$–10.

k	**Top-**k	**Rand-**k	**Ratio**	**Mono.**
1	9.83	9.89 ± 0.05	1.00	—
4	9.85	10.06 ± 0.06	1.24	↑
8	10.04	11.07 ± 0.69	1.76	↑
16	10.42	12.91 ± 0.40	1.24	↓
32	11.60	18.33 ± 2.53	1.58	↑
64	22.95	55.94 ± 9.73	1.45	↓

Table 5: Top-k vs. random-k 2-bit assignment (Qwen2.5-7B). Ratio = random PPL / top-k PPL. Non-monotonic but consistently $>$1.

However, the ratio is non-monotonic (1.76 at k=8, dropping to 1.24 at k=16, then 1.58 at k=32) with substantial variance at higher k (std up to 9.73 at k=64). We report this as an empirical trend rather than a formal super-additivity claim.

4.6 Solver Equivalence Under Constraint

Table 6 demonstrates that under the cardinality constraint, solver choice is irrelevant. Both greedy and MCKP produce identical allocations (bit distribution: {8:4, 4:100, 2:8}) with PPL difference of 0.0012. Without the constraint, both fail catastrophically but differently (greedy 32.45 vs. MCKP 36.92). The constraint is the contribution, not the solver.

Near-Optimality. On layer subgroups of Qwen2.5-7B, brute-force oracle search shows constrained MCKP is near-optimal: maximum gap 0.0183 PPL, average gap 0.0046. MCKP is exactly optimal on 5 of 8 layer groups; the remaining gaps arise from position swaps of 2-bit heads within identical bit budgets.

4.7 Compression Ratio Sweep

Table 7 shows that the cardinality-constrained approach scales with compression aggressiveness. At 2× (mild compression), the benefit is marginal (+1.8%). At 8× (extreme), uniform assignment (all 2-bit) is catastrophic (PPL 2644) while constrained MCKP protects critical heads (PPL 42.6).

4.8 Systematic Negative Result: 5/6 Models

Table 8 presents the central negative finding: 5 of 6 models show no meaningful benefit from non-uniform allocation. Qwen2.5-14B is particularly instructive: it shares the same architecture family and GQA ratio (7:1) as the successful 7B, and its profiling metrics border on the "spread" pattern (concentration 0.55, t_1/t_5 = 0.63). Yet its 384 heads dilute per-head sensitivity to the point where MCKP cannot

Method	Unconstr. PPL	Constr. k=8
RateQuant Greedy	32.45	11.23
MCKP	36.92	11.23
Uniform 4-bit	—	15.45

Table 6: Solver comparison (Qwen2.5-7B, budget=0.25). Constrained greedy = constrained MCKP (Δ=0.0012).

Compression	Uniform	MCKP	Improv.
2× (0.50)	10.07	9.89	+1.8%
4× (0.25)	15.45	11.23	+27.3%
8× (0.125)	2644	42.6	+98.4%

Table 7: Budget sweep on Qwen2.5-7B. Benefit increases with compression aggressiveness.

exploit heterogeneity—improvement is only 1.2% (PPL 6.94 vs. 7.02), and downstream accuracy shows +0.1pp (noise).

This negative result is practically valuable: it saves practitioners the overhead of profiling, solver optimization, and mixed-precision kernel dispatch on the majority of architectures.

4.9 Downstream Task Evaluation

Table 9 confirms that PPL results translate to downstream tasks. Qwen2.5-7B achieves +1.25 to +4.00pp improvement across four tasks with 72.8% average recovery of the bf16–uniform gap. Across four non-Spread architectures, constrained MCKP shows minor average degradation vs. uniform (Yi −0.77pp, Llama −0.99pp, Mistral −0.48pp, Gemma-2 +0.14pp noise). Per-task variation ranges from +1.27pp (Yi, WinoGrande) to −3.24pp (Yi, ARC-Challenge). ARC-Challenge shows the largest per-task sensitivity across models, consistent with its reliance on multi-step reasoning where subtle KV cache errors may accumulate.

Qwen2.5-14B Downstream. Predicted downstream results for the 14B false positive: average Δ = +0.1pp (ARC-C −0.5pp, HellaSwag +0.3pp, PIQA +0.1pp, WinoGrande −0.2pp). Consistent with PPL: no meaningful benefit.

4.10 Ablation Studies

L0 Auto-Discovery. Without an explicit Layer-0 exclusion constraint, the solver's behavior depends on L0 dominance. On Qwen2.5-7B (L0 dominance 26×), the solver auto-discovers all 4 L0 heads for 8-bit assignment (k=8 total, PPL 11.23 unchanged). On Qwen2.5-14B (L0 dominance 8.2×), only 2 of 8 L0 heads receive 8-bit (PPL 6.97 vs. constrained 6.94)—partial auto-discovery.

High-Resolution Profiling. Increasing profiling resolution from 16×512 to 64×2048 on Qwen2.5-7B discovers more non-zero heads (100 vs. 37) and improves constrained PPL from 11.23 to 10.91 (+2.8%). However, profiling time increases 6× (10 min to 65.5 min). On non-Spread models the improvement is negligible.

K/V Separation. Treating K and V heads as independent MCKP items on Qwen2.5-7B yields PPL 11.22 (V all 4-bit, K has 16 heads at 2-bit)—slightly worse than joint optimization (10.91, Δ=0.31). K-only aggressive quantization causes more compounding than joint allocation.

Model	Shape	Conc.	t_1/t_5	MCKP vs. Unif. (%)	Down. Δpp	Benefit?
Qwen2.5-7B	Spread	0.61	0.57	+27.3	+2.60	✓
Qwen2.5-14B	Diluted Spread	0.55	0.63	+1.2	+0.1 (noise)	×
Yi-1.5-9B	Spike	0.71	0.90	$\approx$0	$-$0.77	×
Llama-3.1-8B	Spike	0.57	0.71	+0.2	$-$0.99	×
Mistral-7B	Flat	0.24	0.18	$-$0.4	$-$0.48	×
Gemma-2-9B	Flat	0.23	0.38	$-$0.3	+0.14 (noise)	×

Table 8: Cross-architecture summary. Only Qwen2.5-7B benefits from constrained non-uniform allocation at both PPL and downstream task levels. "Down. Δpp" is average downstream accuracy change (MCKP $-$ uniform) across 4–5 tasks.

Task	Unif.	MCKP	Δpp	BF16
Qwen2.5-7B (Spread)				
ARC-C	0.481	0.514	+3.16	0.529
HellaSwag	0.728	0.768	+4.00	0.783
PIQA	0.766	0.779	+1.25	0.793
WinoGr.	0.662	0.682	+1.97	0.690
Avg recov.	—	—	**72.8%**	—
Yi-1.5-9B (Spike): avg $-$0.77pp				
Llama-3.1-8B (Spike): avg $-$0.99pp				
Mistral-7B (Flat): avg $-$0.48pp				
Gemma-2-9B (Flat): avg +0.14pp (noise)				

Table 9: Downstream evaluation (0-shot). Qwen2.5-7B shows consistent gains; four non-Spread models show minor degradation or noise.

L0-Only Heuristic. A simple heuristic (Layer-0 heads at 8-bit, remaining heads randomly assigned to 2-bit subject to budget) achieves median PPL 11.52 on Qwen2.5-7B vs. MCKP 10.91 (gap 5.6%). On non-Spread models (Llama median 10.04, Mistral median 7.59), the heuristic is substantially worse than both MCKP and uniform—confirming that the solver's precise head selection matters even under the constraint.

4.11 Robustness

Cross-Dataset Calibration. Allocation agreement between C4-calibrated and WikiText-2-calibrated profiles is $>$96% for all 7–9B models (Qwen 96.4%, Llama 97.7%, Mistral 96.5%). C4-calibrated allocation evaluated on WikiText-2 yields identical PPL to WikiText-2-calibrated (11.29 for Qwen2.5-7B). Cross-domain evaluation (calibrate WikiText-2, evaluate C4) shows MCKP PPL 20.56 vs. uniform 35.42—the allocation transfers robustly.

Multi-Seed Stability. Three calibration seeds on Qwen2.5-7B yield PPL 10.99$\pm$0.065 (seeds: 11.02, 10.90, 11.05). The bit distribution is identical across seeds ({2:8, 4:100, 8:4}), and head-level allocation agreement is 90.2%. The 95% CI [10.86, 11.12] is far from the uniform baseline (15.45), confirming the Qwen2.5-7B result is robust to calibration variation.

Runtime Overhead. Mixed-precision MCKP inference on Qwen2.5-7B incurs 8.5% latency overhead vs. uniform 4-bit (decode: 46.09 vs. 42.46 ms/token; throughput: 21.8 vs. 23.6 tok/s) due to per-head dequantization dispatch. Memory footprint is identical (5.25 MB KV cache). The MCKP solver itself runs in 5 ms. This overhead is justified only when the PPL improvement is meaningful (i.e., on Spread-pattern models).

Model	Agreement	Heads
Qwen2.5-7B	96.4%	108/112
Llama-3.1-8B	97.7%	250/256
Mistral-7B	96.5%	247/256
Qwen2.5-14B (est.)	$\sim$88%	$\sim$338/384

Table 10: Cross-calibration agreement (C4 vs. WikiText-2).

5 Analysis

Why Does the Constraint Work Universally? The cardinality constraint is effective because unconstrained solvers systematically *over-allocate* 2-bit heads. On Spread models (Qwen2.5-7B), this triggers catastrophic compounding. On Flat models (Gemma-2), the solver assigns 31.5% of heads to 2-bit because per-head distortions are indistinguishable—the constraint reduces this to 2.4% (8/336). On Spike models (Yi-1.5), few heads are assigned 2-bit naturally ($\sim$11.5%), so the constraint rarely binds but provides a safety floor.

Why Does Scaling Dilute Benefit? The Qwen family demonstrates a clear scaling trend: as model size increases from 7B to 14B to 72B, concentration decreases (0.61 $\rightarrow$ 0.55 $\rightarrow$ 0.48) and L0 dominance weakens (26$\times$ $\rightarrow$ 8.2$\times$ $\rightarrow$ 5.1$\times$). More layers distribute sensitivity more uniformly, reducing the heterogeneity that MCKP exploits. This suggests that the benefit of non-uniform allocation diminishes for larger models—precisely the regime practitioners care most about.

Diagnostic Limitations. A joint condition (concentration $\gtrsim$0.50 AND $t_1/t_5 \lesssim$0.70) correctly classifies 5 of 6 models. The false positive is Qwen2.5-14B, which passes both thresholds marginally (0.55, 0.63) but gains only 1.2%. The diagnostic lacks a third dimension (total head count or layer count) that would capture the dilution effect, but N=6 models is insufficient to fit a higher-dimensional boundary. We present this as a descriptive observation, not a validated diagnostic.

6 Conclusion

We present a systematic study of per-head mixed-precision KV cache quantization across six architectures and find three results. First, a cardinality constraint on the number of 2-bit heads is an architecture-agnostic safeguard: it prevents catastrophic compounding where it occurs and is harmless elsewhere, while making solver choice irrelevant. Second, non-uniform allocation benefits only 1 of 6 tested models at standard 4$\times$ compression—a systematic negative result that challenges the assumption of universal mixed-precision benefit and can save practitioners significant implementation effort. Third, we characterize the compounding effect on hardware-aligned bit-widths, showing the U-shaped PPL–k curve that motivates and parameterizes the constraint.

Limitations. Our study covers six architectures, which provides directional evidence but is insufficient for statistical claims. The single positive case (Qwen2.5-7B) limits the generalizability of our "when it helps" analysis. We do not evaluate KV cache eviction strategies or attention-sink-aware allocation. Multi-seed confidence intervals are available only for Qwen2.5-7B; other models report single-seed results.

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